

No.	Description	Type	Alternative Part	PCB Marking	Units	Remarks
Jupiter ACE PCB i2.0						
1	Z80 Series, NMOS / CMOS, 4 MHz, 4.5 V to 5 V, DIP40	Z80A		Z0	1	CPU_CLK = 3.25MHz
2	4K x 8bit EPROM; DIP24W	27C32		Z1, Z2	2	modified for EPROM IC's; 200
3	1K x 4bit SRAM; DIP18	2114-2	2114-3	Z3, Z4	2	Variable, dictionary RAM; 200-300ns; in matching pair
4	1K x 4bit SRAM; DIP18	2114-2	2114-3	Z5, Z6	2	Video RAM; 200-300ns; in matching pair
5	1K x 4bit SRAM; DIP18	2114-2	2114-3	Z7, Z8	2	CHR\$ RAM; 200-300ns; in matching pair
6	Dual 4-stage Binary Counter; DIP14	74LS393		Z9-Z11	3	
7	Parallel-Load 8-bit Shift Register; DIP16	74LS166		Z12	1	Marked as Z28 in Bodo's schem.
8	1-of-8 Decoder/Demultiplexer; DIP16	74LS138		Z13	1	Marked as Z29 in Bodo's schem.
7	Hex 3-state Buffer; DIP16	74LS367		Z14-Z18	5	Keyboard, VRAM & CHR\$ RAM buffers
8	Quad 2-input NAND Gate; DIP14	74LS00		Z19	1	
9	Quad 2-input AND Gate; DIP14	74LS08		Z20	1	
10	Triple 3-input AND Gate; DIP14	74LS11		Z21, Z22	2	
11	Quad 2-input ExOR Gate; DIP14	74LS86		Z23	1	
12	Quad 2-input NOR Gate; DIP14	74LS02		Z24, Z25	2	
13	Quad 2-input OR Gate; DIP14	74LS32		Z26	1	
14	Dual D-type Positive-Edge-Triggered Flip-Flop; DIP14	74LS74		Z27	1	
15						
16	Positive Voltage Regulator 5V, 1A; TO-220	TSR1-2450	LM7805	U27	1	DC-DC converter recommended
17						
18	Crystal oscillator; HC-49	6.5MHz	6.5536MHz	X1	1	6.5MHz RC oscillator might be used
19						
20	NPN General Purpose Transistor; TO-92	2N3904		Q1	1	6.5MHz generator
21	NPN Fast Switching Transistor; TO-92	MPS2369	ZTX313	Q2	1	CPU_CLK driver
22	NPN General Purpose Transistor; TO-92	2N3904	BC546-BC550	Q3	1	Q3 (BC183) in 4000 & Bodo's schem.
23						
24	General Purpose Diode; DO-35	1N4148		D1-D13	13	D2 in 4000 & Bodo's schem.; D12 & D13 on soldering side (if needed)
25						
26	Ceramic capacitor, 50V, LS=5.08mm	39pF		C1	1	
27	Ceramic capacitor, 50V, LS=5.08mm	100pF		C2	1	
28	Ceramic capacitor, 50V, LS=5.08mm	47nF		C3, C6	2	C3 (EAR_IN) 1uF recommended
29	Ceramic capacitor, 50V, LS=5.08mm	47pF		C4, C5	2	C5 (MIC-SPK) not fitted
30	Electrolytic capacitor, 25V, dia. 5mm LS=2.5mm	1uF/25V		C7	1	/RST line
31	Ceramic capacitor, 50V, LS=5.08mm	2.2nF		C8	1	/WAIT line
32	Electrolytic capacitor, 25, dia. 6.3mm LS=3.5mm	100uF/25V		C9, C12	2	
33	Ceramic capacitor, 50V, LS=5.08mm	100nF	47nF to 220nF	C10, C13-C19	8	
34						
35	Resistor 1/4W	4k7		R1	1	
36	Resistor 1/4W	47k		R2	1	
37	Resistor 1/4W	1k0		R3, R4, R6, R7, R11-R23, R28-R30	20	R29 & R30 on soldering side (if needed)
38	Resistor 1/4W	12k		R5	1	
39	Resistor 1/4W	270R		R8	1	
40	Resistor 1/4W	22R		R9	1	Speaker driver; 220R in 4000 & Bobo's schem.
41	Resistor 1/4W	10k		R10, R24	1	R24 (/BUSRQ) in Bodo's schem.
42	Resistor 1/4W	220k		R25	1	/RST line
43	Resistor 1/4W	330R	470R (TBC)	R26	1	
44	Resistor 1/4W	33R		R27	1	
45						
46	Resistor network; DIP16	8 x 1k0	16 x 1k0 resistor	RN1, RN2	2	1k0 resistors to be used
47						
48	Speaker (220R); dia. ???	220R		LS1	1	
49						
50	3.5mm socket	FC68130	JYO-36A	JP1, JP2, JP4	3	
51						
52	IC socket DIP14				12	Logic ICs
53	IC socket DIP16				7	Logic ICs
54	IC socket DIP18				6	SRAM
55	IC socket DIP24W				2	ROM
56	IC socket DIP40				1	CPU
57						
58	RF Modulator E36	UM1233		RF	1	Empty case only
59						
60	RF Modulator					
61	NPN General Purpose Transistor; TO-92	2N3904	2N2222	RF-Q1	1	Any NPN type transistor should work.
62						
63	Electrolytic capacitor, 16V, dia. 6.3mm LS=3.5mm	100uF/16V		RF-C1, RF-C3	2	Output capacitor needed only
64	Ceramic capacitor, 50V, LS=5.08mm	100nF		RF-C2	1	Not essential
65						
66	Resistor 1/4W	2k2	1k5 to 3k3	RF-R1	1	
67	Resistor 1/4W	75R	68R-100R	RF-R2	1	150R trimmer would be better
68	Resistor 1/4W	10R		RF-R3	1	Not essential